Supplementary Material

When Simple Is Enough: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction

This document holds the material that supports the manuscript without being part of its argument: the architecture chosen by the inner loop in every outer fold, and the detail of the screening, docking and ADMET stage, whose results are hypotheses for experimental testing rather than conclusions of the benchmark. Every value is generated by the analysis pipeline and read from the same files as the manuscript.

Benchmark detail

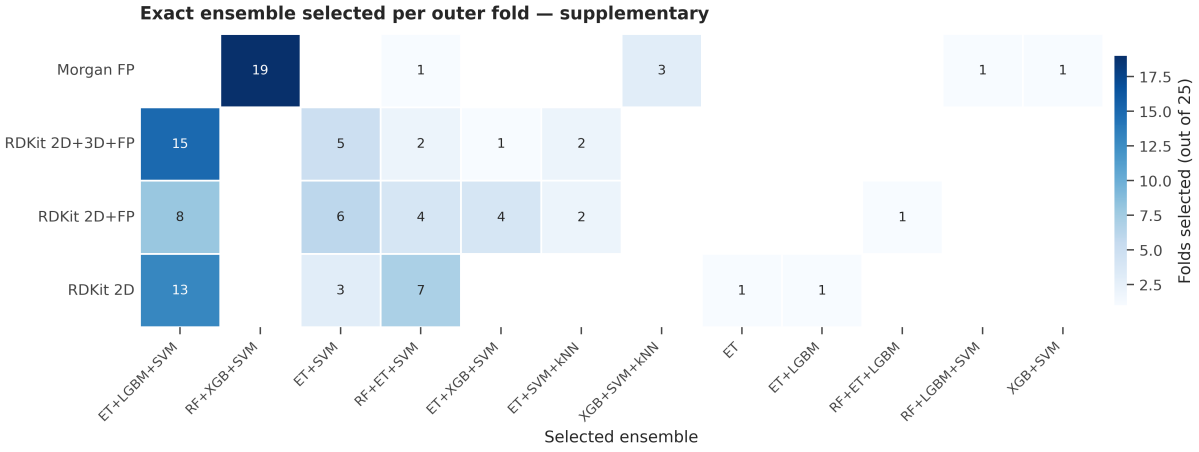


Figure S1: Stacking ensemble selected by the inner loop in each of the 25 outer folds, for each molecular representation. Cell values are the number of outer folds in which that exact combination of base learners (RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, k-nearest neighbors) was chosen. No combination was chosen consistently, which shows that the inner search does not converge on a single architecture.

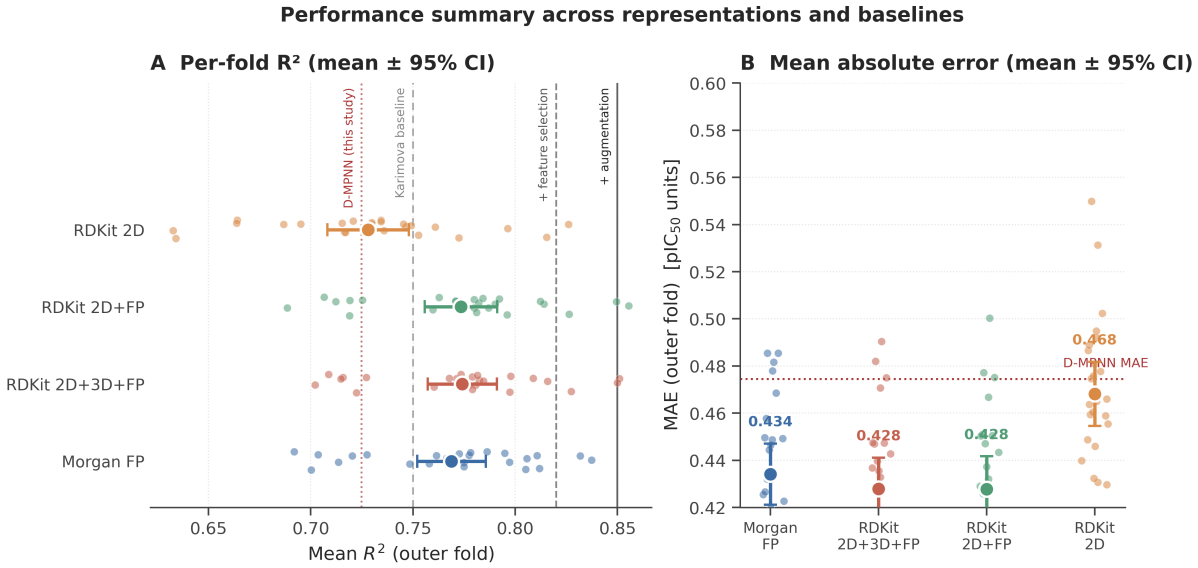


Figure S2: Per-fold performance of the four representations over the 25 outer folds. (A) R^2 of each fold (faint points), with the mean and its 95% confidence interval; (B) the same for the mean absolute error. The vertical lines mark the D-MPNN of this study and the three values reported by the original study, which come from a single hold-out split of a pipeline with the leaks quantified in Section 3.1 of the manuscript and are therefore shown for reference only, not as a paired comparison.

Table S1: Every pairwise comparison between the four molecular representations over the 25 shared outer folds, in both metrics. Differences are A minus B, the interval is the 90% confidence interval used by the equivalence test, p is the Nadeau–Bengio corrected paired t-test, p_{Holm} its step-down adjustment within the six comparisons of each metric, and p_{equiv} the two one-sided tests against the ± 0.05 margin.

Metric	Comparison	Difference	90% CI	p	p_{Holm}	p_{equiv}
R^2	Morgan vs RDKit 2D	+0.0408	[+0.021, +0.061]	0.0020	0.0081	0.2201
R^2	Morgan vs RDKit 2D+FP	−0.0047	[−0.016, +0.007]	0.5061	1.0000	<0.0001
R^2	Morgan vs RDKit 2D+3D+FP	−0.0052	[−0.016, +0.005]	0.4006	1.0000	<0.0001
R^2	RDKit 2D vs RDKit 2D+FP	−0.0454	[−0.063, −0.028]	0.0002	0.0009	0.3288
R^2	RDKit 2D vs RDKit 2D+3D+FP	−0.0460	[−0.063, −0.029]	0.0001	0.0008	0.3489
R^2	RDKit 2D+FP vs RDKit 2D+3D+FP	−0.0006	[−0.005, +0.004]	0.8243	1.0000	<0.0001
MAE	Morgan vs RDKit 2D	−0.0340	[−0.051, −0.017]	0.0025	0.0099	0.0625
MAE	Morgan vs RDKit 2D+FP	+0.0065	[−0.003, +0.016]	0.2739	0.7689	<0.0001
MAE	Morgan vs RDKit 2D+3D+FP	+0.0063	[−0.003, +0.016]	0.2563	0.7689	<0.0001
MAE	RDKit 2D vs RDKit 2D+FP	+0.0405	[+0.026, +0.054]	<0.0001	0.0002	0.1273
MAE	RDKit 2D vs RDKit 2D+3D+FP	+0.0403	[+0.028, +0.053]	<0.0001	<0.0001	0.0980
MAE	RDKit 2D+FP vs RDKit 2D+3D+FP	−0.0001	[−0.005, +0.005]	0.9664	0.9664	<0.0001

Table S2: Composition of the stacking ensembles chosen by the inner loop, per representation. Each percentage is the share of the 25 outer folds in which that base learner entered the selected ensemble; stability is the share taken by the single most frequent exact combination. RF, random forest; ET, extra trees; LGBM, LightGBM; XGB, XGBoost; SVM, support vector regression; kNN, k-nearest neighbors.

Representation	RF	ET	LGBM	XGB	SVM	kNN	**Most frequent**	**Stability**
Morgan FP	84%	4%	4%	92%	100%	12%	RF+XGB+SVM	76.0%
RDKit 2D+3D+FP	8%	100%	60%	4%	100%	8%	ET+LGBM+SVM	60.0%
RDKit 2D+FP	20%	100%	36%	16%	96%	8%	ET+LGBM+SVM	32.0%
RDKit 2D	28%	100%	56%	0%	92%	0%	ET+LGBM+SVM	52.0%

Table S3: The four y-randomization tests of Rücker et al., each with 100 randomizations scored by 5-fold cross-validation. Scope "architecture" randomizes the stacking ensemble on all 2048 bits; scope "procedure" repeats the complete model-building procedure, including the out-of-fold permutation-importance selection that produces the deployed screening model, inside every randomization. The randomization is either a permutation of the response or a replacement of the descriptors by random ones. Z is the distance from the real model to the mean of the randomized ones in units of their standard deviation; p cannot fall below $1/(100+1)$.

Scope	**Randomization**	R^2 (real)	R^2 (randomized)	Z	p
Architecture	Response permuted	0.7787	-0.0136 ± 0.0084	93.9	0.0099
Architecture	Random descriptors	0.7787	-0.0136 ± 0.0083	95.7	0.0099
Procedure	Response permuted	0.7722	-0.0365 ± 0.0169	47.8	0.0099
Procedure	Random descriptors	0.7722	-0.0959 ± 0.0322	27.0	0.0099

Table S4: Each additional neural model against each classical representation over the 25 shared outer folds. Differences are the neural model minus the representation, so a negative value in R^2 means the neural model scored lower. p_{Holm} is adjusted within the four representations of each model and metric.

Metric	Comparison	Difference	90% CI	p_{raw}	p_{Holm}	p_{equiv}
R^2	MLP on ECFP4 vs Morgan	−0.0598	[−0.104, −0.015]	0.0300	0.0965	0.6458
R^2	MLP on ECFP4 vs RDKit 2D	−0.0190	[−0.069, +0.031]	0.5193	0.5193	0.1492
R^2	MLP on ECFP4 vs RDKit 2D+FP	−0.0645	[−0.110, −0.019]	0.0241	0.0965	0.7029
R^2	MLP on ECFP4 vs RDKit 2D+3D+FP	−0.0651	[−0.111, −0.019]	0.0246	0.0965	0.7080
R^2	Ensemble of 5 D-MPNN vs Morgan	−0.0279	[−0.051, −0.005]	0.0505	0.1622	0.0582
R^2	Ensemble of 5 D-MPNN vs RDKit 2D	+0.0128	[−0.016, +0.042]	0.4598	0.4598	0.0199
R^2	Ensemble of 5 D-MPNN vs RDKit 2D+FP	−0.0326	[−0.059, −0.006]	0.0468	0.1622	0.1365
R^2	Ensemble of 5 D-MPNN vs RDKit 2D+3D+FP	−0.0332	[−0.059, −0.007]	0.0406	0.1622	0.1413
R^2	D-MPNN with RDKit 2D descriptors vs Morgan	−0.0681	[−0.103, −0.033]	0.0030	0.0121	0.8049
R^2	D-MPNN with RDKit 2D descriptors vs RDKit 2D	−0.0273	[−0.068, +0.014]	0.2657	0.2657	0.1764
R^2	D-MPNN with RDKit 2D descriptors vs RDKit 2D+FP	−0.0727	[−0.112, −0.034]	0.0040	0.0121	0.8353
R^2	D-MPNN with RDKit 2D descriptors vs RDKit 2D+3D+FP	−0.0733	[−0.112, −0.035]	0.0031	0.0121	0.8464
R^2	D-MPNN initialized from CheMeleon vs Morgan	−0.0074	[−0.033, +0.018]	0.6262	1.0000	0.0044
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D	+0.0334	[−0.002, +0.069]	0.1216	0.4866	0.2164
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	−0.0120	[−0.040, +0.016]	0.4753	1.0000	0.0155
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	−0.0126	[−0.040, +0.015]	0.4357	1.0000	0.0136
MAE	MLP on ECFP4 vs Morgan	+0.0648	[+0.032, +0.097]	0.0024	0.0058	0.7770
MAE	MLP on ECFP4 vs RDKit 2D	+0.0308	[−0.006, +0.067]	0.1649	0.1649	0.1896
MAE	MLP on ECFP4 vs RDKit 2D+FP	+0.0712	[+0.037, +0.106]	0.0017	0.0058	0.8482
MAE	MLP on ECFP4 vs RDKit 2D+3D+FP	+0.0711	[+0.037, +0.105]	0.0015	0.0058	0.8517
MAE	Ensemble of 5 D-MPNN vs Morgan	+0.0258	[+0.002, +0.050]	0.0775	0.1550	0.0478
MAE	Ensemble of 5 D-MPNN vs RDKit 2D	−0.0082	[−0.030, +0.014]	0.5262	0.5262	0.0017
MAE	Ensemble of 5 D-MPNN vs RDKit 2D+FP	+0.0322	[+0.008, +0.056]	0.0305	0.0933	0.1083
MAE	Ensemble of 5 D-MPNN vs RDKit 2D+3D+FP	+0.0321	[+0.009, +0.055]	0.0233	0.0933	0.0946
MAE	D-MPNN with RDKit 2D descriptors vs Morgan	+0.0608	[+0.028, +0.093]	0.0037	0.0073	0.7136
MAE	D-MPNN with RDKit 2D descriptors vs RDKit 2D	+0.0268	[−0.002, +0.056]	0.1250	0.1250	0.0906
MAE	D-MPNN with RDKit 2D descriptors vs RDKit 2D+FP	+0.0672	[+0.035, +0.100]	0.0018	0.0053	0.8122
MAE	D-MPNN with RDKit 2D descriptors vs RDKit 2D+3D+FP	+0.0671	[+0.036, +0.098]	0.0012	0.0048	0.8209
MAE	D-MPNN initialized from CheMeleon vs Morgan	+0.0075	[−0.020, +0.035]	0.6431	1.0000	0.0067
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D	−0.0265	[−0.055, +0.002]	0.1291	0.5164	0.0886
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	+0.0139	[−0.014, +0.042]	0.3964	1.0000	0.0174
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	+0.0138	[−0.012, +0.040]	0.3729	1.0000	0.0128

Table S5: Comparisons under scaffold-grouped folds: every neural model against every classical representation, and the classical representations against each other. The unadjusted p is given beside the adjusted one because the 90% intervals quoted in the manuscript are not corrected for multiplicity, which is why an interval can exclude zero while the adjusted p is far from significance.

Metric	**Comparison**	**Difference**	**90% CI**	p_{raw}	p_{Holm}	p_{equiv}
R^2	D-MPNN, default vs Morgan	−0.0729	[−0.144, −0.002]	0.0919	0.3676	0.7070
R^2	D-MPNN, default vs RDKit 2D	+0.0291	[−0.045, +0.103]	0.5067	0.7607	0.3161
R^2	D-MPNN, default vs RDKit 2D+FP	−0.0495	[−0.125, +0.026]	0.2737	0.7607	0.4954
R^2	D-MPNN, default vs RDKit 2D+3D+FP	−0.0528	[−0.130, +0.024]	0.2536	0.7607	0.5243
R^2	DNN baseline vs Morgan	−0.1053	[−0.215, +0.004]	0.1118	0.4470	0.8028
R^2	DNN baseline vs RDKit 2D	−0.0033	[−0.114, +0.107]	0.9594	0.9594	0.2380
R^2	DNN baseline vs RDKit 2D+FP	−0.0819	[−0.200, +0.036]	0.2453	0.6965	0.6765
R^2	DNN baseline vs RDKit 2D+3D+FP	−0.0852	[−0.204, +0.034]	0.2322	0.6965	0.6914
R^2	DNN with selection and augmentation vs Morgan	−0.0837	[−0.159, −0.008]	0.0703	0.2812	0.7735
R^2	DNN with selection and augmentation vs RDKit 2D	+0.0183	[−0.072, +0.109]	0.7328	0.7328	0.2777
R^2	DNN with selection and augmentation vs RDKit 2D+FP	−0.0603	[−0.147, +0.026]	0.2445	0.6514	0.5796
R^2	DNN with selection and augmentation vs RDKit 2D+3D+FP	−0.0636	[−0.149, +0.022]	0.2171	0.6514	0.6054
R^2	Ensemble of D-MPNN vs Morgan	−0.0392	[−0.106, +0.028]	0.3259	0.9778	0.3919
R^2	Ensemble of D-MPNN vs RDKit 2D	+0.0629	[−0.013, +0.139]	0.1684	0.6736	0.6131
R^2	Ensemble of D-MPNN vs RDKit 2D+FP	−0.0157	[−0.092, +0.060]	0.7272	1.0000	0.2242
R^2	Ensemble of D-MPNN vs RDKit 2D+3D+FP	−0.0190	[−0.096, +0.058]	0.6773	1.0000	0.2493
R^2	D-MPNN initialized from CheMeleon vs Morgan	+0.0065	[−0.038, +0.051]	0.8035	0.9247	0.0538
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D	+0.1086	[+0.051, +0.166]	0.0035	0.0142	0.9531
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	+0.0300	[−0.019, +0.079]	0.3082	0.9247	0.2471
R^2	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	+0.0267	[−0.025, +0.078]	0.3851	0.9247	0.2238
MAE	D-MPNN, default vs Morgan	+0.0486	[−0.002, +0.100]	0.1160	0.4638	0.4820
MAE	D-MPNN, default vs RDKit 2D	−0.0260	[−0.081, +0.029]	0.4298	0.8935	0.2332
MAE	D-MPNN, default vs RDKit 2D+FP	+0.0335	[−0.023, +0.090]	0.3176	0.8935	0.3096
MAE	D-MPNN, default vs RDKit 2D+3D+FP	+0.0348	[−0.021, +0.091]	0.2978	0.8935	0.3228
MAE	DNN baseline vs Morgan	+0.0159	[−0.030, +0.061]	0.5565	1.0000	0.1058
MAE	DNN baseline vs RDKit 2D	−0.0588	[−0.115, −0.002]	0.0885	0.3541	0.6037
MAE	DNN baseline vs RDKit 2D+FP	+0.0007	[−0.056, +0.058]	0.9835	1.0000	0.0763
MAE	DNN baseline vs RDKit 2D+3D+FP	+0.0020	[−0.054, +0.058]	0.9523	1.0000	0.0790
MAE	DNN with selection and augmentation vs Morgan	+0.0125	[−0.050, +0.075]	0.7378	1.0000	0.1587
MAE	DNN with selection and augmentation vs RDKit 2D	−0.0622	[−0.132, +0.007]	0.1383	0.5532	0.6170
MAE	DNN with selection and augmentation vs RDKit 2D+FP	−0.0027	[−0.074, +0.068]	0.9485	1.0000	0.1328
MAE	DNN with selection and augmentation vs RDKit 2D+3D+FP	−0.0014	[−0.072, +0.069]	0.9729	1.0000	0.1250
MAE	Ensemble of D-MPNN vs Morgan	+0.0244	[−0.020, +0.069]	0.3561	1.0000	0.1659
MAE	Ensemble of D-MPNN vs RDKit 2D	−0.0503	[−0.106, +0.005]	0.1357	0.5427	0.5037
MAE	Ensemble of D-MPNN vs RDKit 2D+FP	+0.0092	[−0.048, +0.067]	0.7866	1.0000	0.1181
MAE	Ensemble of D-MPNN vs RDKit 2D+3D+FP	+0.0105	[−0.047, +0.068]	0.7584	1.0000	0.1264
MAE	D-MPNN initialized from CheMeleon vs Morgan	+0.0005	[−0.030, +0.032]	0.9775	1.0000	0.0058
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D	−0.0741	[−0.109, −0.039]	0.0014	0.0056	0.8742
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+FP	−0.0146	[−0.052, +0.022]	0.5033	1.0000	0.0570
MAE	D-MPNN initialized from CheMeleon vs RDKit 2D+3D+FP	−0.0133	[−0.052, +0.026]	0.5632	1.0000	0.0603
R^2	Morgan vs RDKit 2D	+0.1020	—	0.0002	0.0015	—
R^2	Morgan vs RDKit 2D+FP	+0.0235	—	0.3332	0.9997	—
R^2	Morgan vs RDKit 2D+3D+FP	+0.0202	—	0.4151	0.9997	—
R^2	RDKit 2D vs RDKit 2D+FP	−0.0786	—	0.0028	0.0139	—
R^2	RDKit 2D vs RDKit 2D+3D+FP	−0.0819	—	0.0031	0.0139	—
R^2	RDKit 2D+FP vs RDKit 2D+3D+FP	−0.0033	—	0.6539	0.9997	—

Virtual screening, docking and ADMET

Table S6: Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics. LE, ligand efficiency; BEI, binding efficiency index; LLE, lipophilic ligand efficiency; SEI, surface efficiency index.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Trimetrexate Glucuronate	7.9272	0.40	21.46	5.19	6.74
2	Methotrexate	7.4448	0.31	16.38	7.18	3.54
3	Triamterene	7.0081	0.51	27.67	6.17	5.41
4	Pralatrexate	7.0034	0.27	14.67	6.02	3.38
5	Pyrimethamine	6.5705	0.53	26.42	4.05	8.44
6	Folic Acid	6.3379	0.27	14.36	6.38	2.97
7	Naldemedine Tosylate	6.0892	0.20	10.67	2.61	4.31
8	Bendamustine Hydrochloride	6.0850	0.36	16.98	2.82	10.43
9	Levomefolate Calcium	6.0732	0.25	13.22	6.33	3.00
10	Letermovir	5.9681	0.20	10.42	0.26	7.67

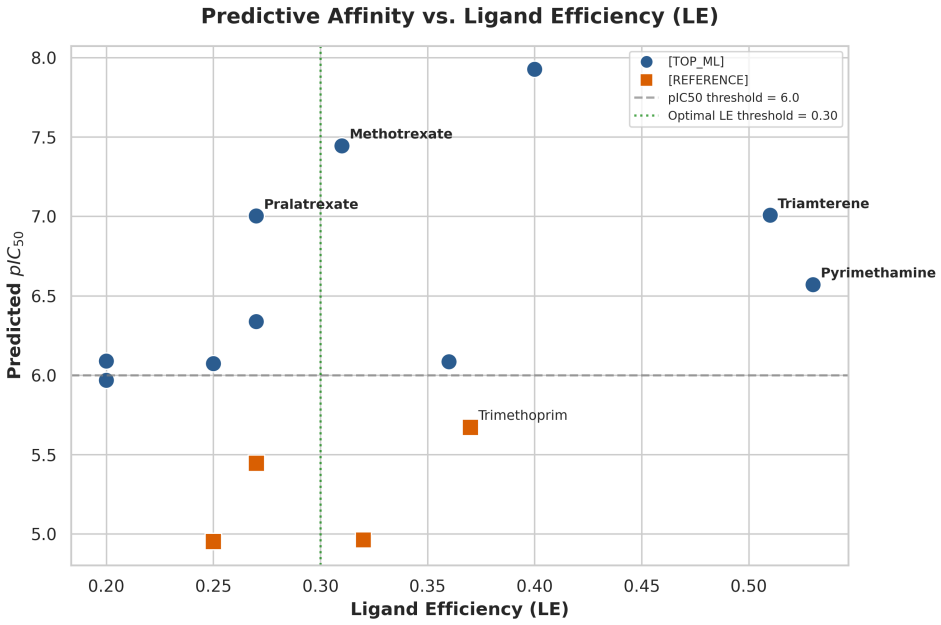


Figure S3: Predicted pIC_{50} versus ligand efficiency for the candidates advanced to docking. Blue: the ten top-ranked compounds by predicted potency; orange: reference compounds from the original study. The dashed line marks $pIC_{50} = 6.0$ and the dotted line LE = 0.30.

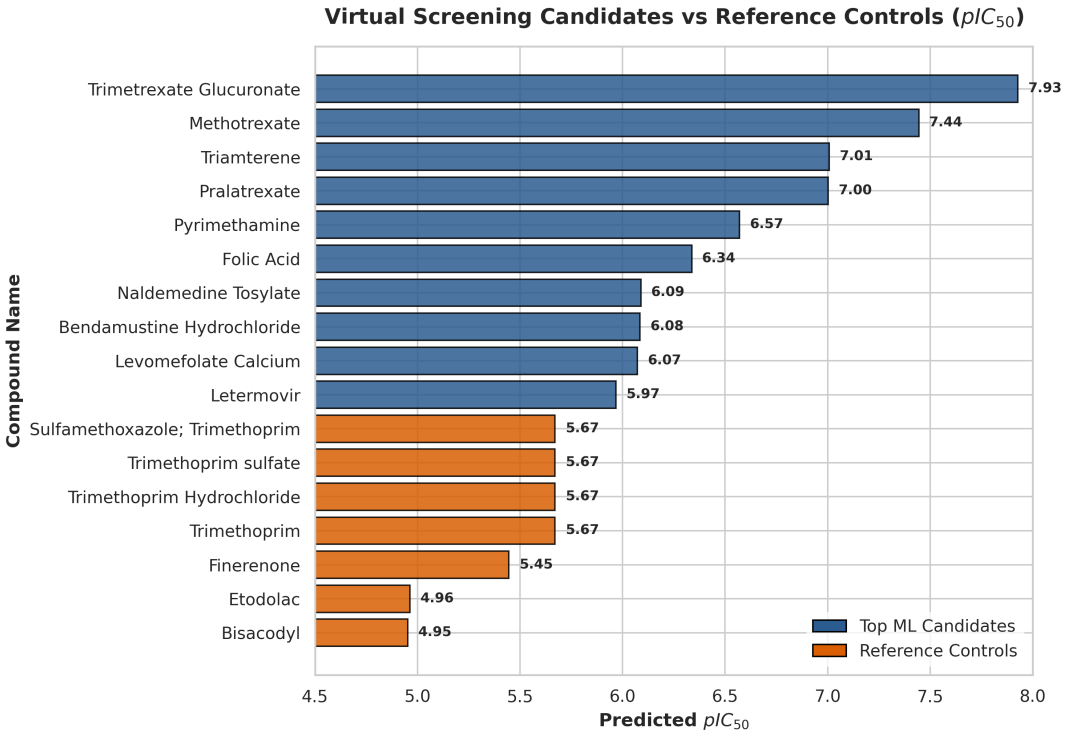


Figure S4: Predicted pIC_{50} of the screening candidates next to the reference controls of the original study. Compounds that belong to the training partition are fitted values and not predictions on unseen molecules.

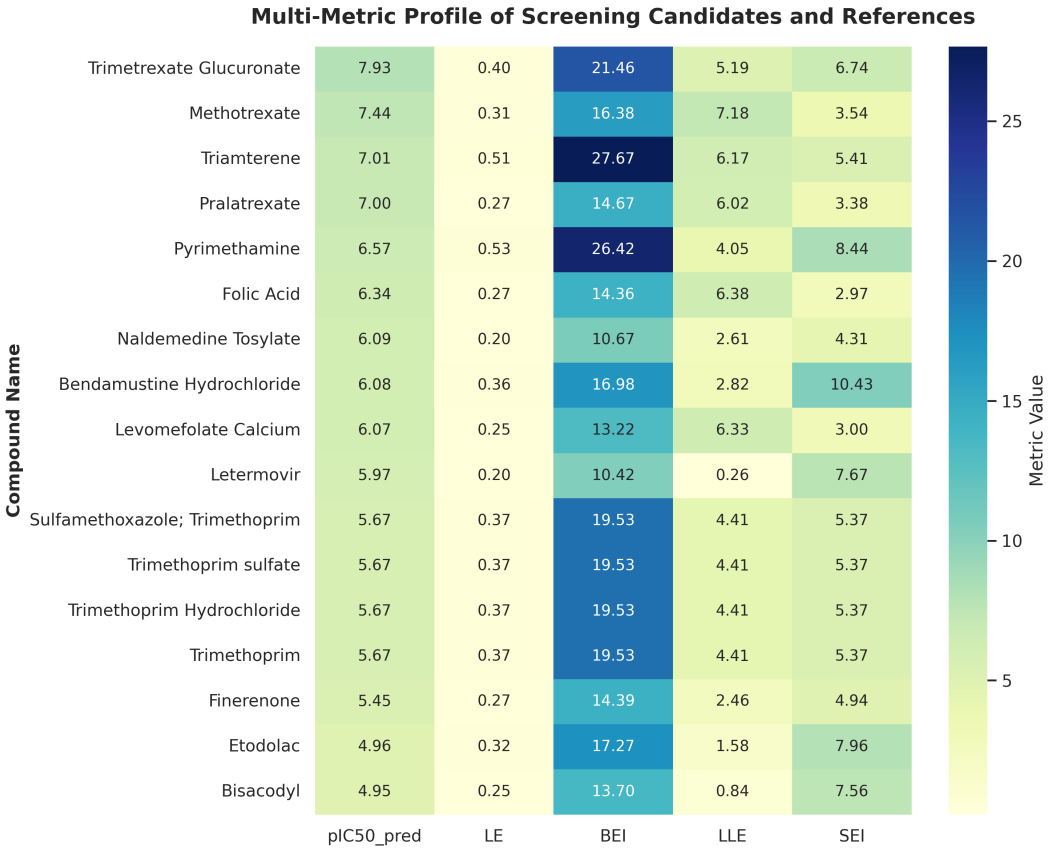


Figure S5: Efficiency metrics of the screening candidates and the reference controls. Each metric is scaled across compounds, so the colour shows a compound's standing within this set and not an absolute quality.

Table S7: Predicted ADMET profile of the 4 consensus hits and of the known inhibitors pyrimethamine and trimethoprim. Structures that appear under several entries are listed once. hERG prob. is the predicted probability of channel blockade (flag above 0.5) and Caco-2 $\log P_{app}$ the predicted permeability (oral if above -5.15). The predictions come from random forests trained on public data whose accuracy was not evaluated here, and some of these compounds are in their training sets, so the profile is a coarse prioritization and not a safety assessment.

Compound	**Pred. pIC$_{50}$**	**ΔG (kcal/mol)**	**hERG prob.**	**Caco-2 $\log P_{app}$**	**Profile**
Trimetrexate Glucuronate	7.93	−8.76	0.40	−4.86	Oral
Pralatrexate	7.00	−8.68	0.31	−5.41	Parenteral
Methotrexate	7.44	−8.66	0.40	−5.87	Parenteral
Triamterene	7.01	−8.07	0.80	−4.66	hERG flag
Pyrimethamine	6.57	−7.62	0.12	−4.42	Oral
Trimethoprim	5.67	−7.56	0.14	−4.77	Oral

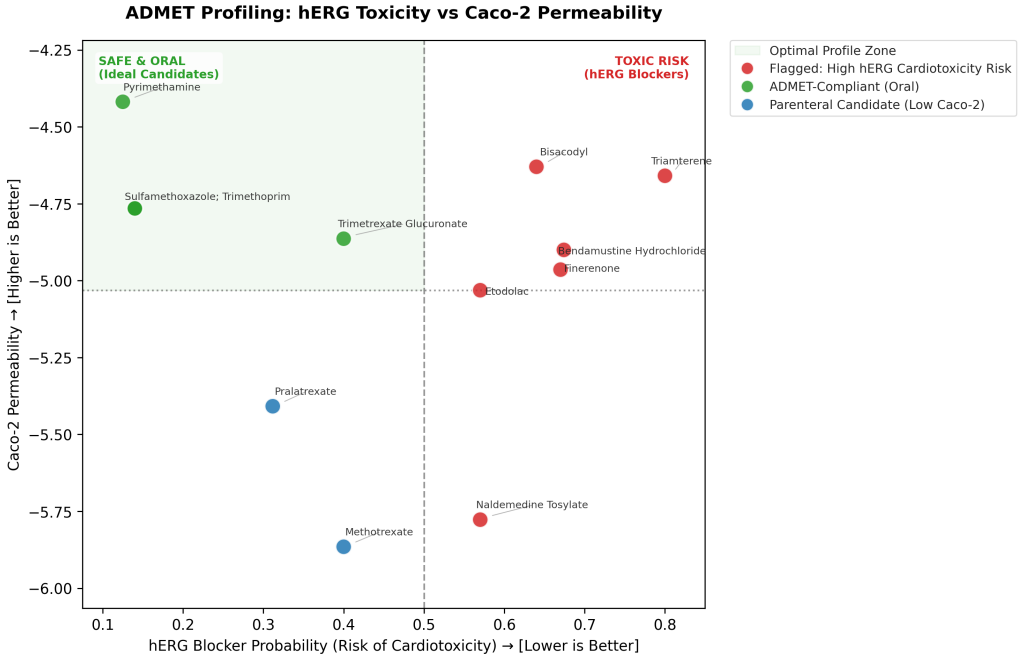


Figure S6: Predicted hERG blocking probability versus predicted Caco-2 permeability for the docked candidates that passed the Lipinski criteria. The dashed lines mark the hERG threshold and the Caco-2 threshold; the shaded quadrant combines low hERG risk with permeability classified as oral.